EE6320 RF Integrated Circuits
Project: Mixer Design

Mixer Performance Summary Table

	Design Metric	Performance	Specification
Conversion Gain	Gain in the specified band [$f_{RF} = f_{LO}$]	20.26 dB	>20 *dB*
	Gain in the specified band [$f_{RF} = f_{LO} + 10MHz$]	20.21 dB	>20 *dB*
	Peak Gain flatness in specified band [Max-Min Gain]	0.05 dB	-
	3*dB* RF Bandwidth [From the plot of $f_{RF} = f_{LO}$]	101.85 MHz	-
Noise Figure	Integrated SSB Noise Figure for $f_{LO} = 3.5\ GHz$	10.643 dB	≤ 12 dB
	RF Frequency Band over which SSB NF<12dB fLO=3.5GHz	1k - 1.1M Hz	≤ 12 dB
Linearity - IIP_2	Input power used for extrapolation	-50 dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-29.75 dBm	-
	Power of IM_2 Tone at output (at chosen input power)	-148.21 dBm	-
	Extrapolated IIP_2 (Mismatch of 0.5%)	68.46 dBm	≥ +20*dBm*
Linearity - IIP_3	Input power used for extrapolation	-50 dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-29.75 dBm	-
	Power of IM_3 Tone at output (at chosen input power)	-121.74 dBm	-
	Extrapolated IIP_3	-4.0029 dBm	≥ *-5 dBm*
Power	Mixer DC power consumption [Excluding Bias] (Total = 2*Individual)	Total = 0.782 mW Each = 0.3911 mW	Minimize
	Bias circuit power consumption	28.8 micro W	Minimize
Other	Sum of all resistances [excluding bias]	2.15*2*2 = 8.6k	-
	Sum of biasing resistances	2*2*10k = 80K	-
	Sum of all capacitances [Including AC coupling] 10p AC coupling for each RF, 723f at each load	10p*2*2 + 723f*2*2 = 42.89pf	-
	Sum of all inductances	0	

	Load chosen (each R_load)	2.15k ohm	-

	Differential Mixer Input Capacitance (C_gs Caps)Each RF fet)	29.92f F	-
	Simulator Used	ADE L, Spectre	-

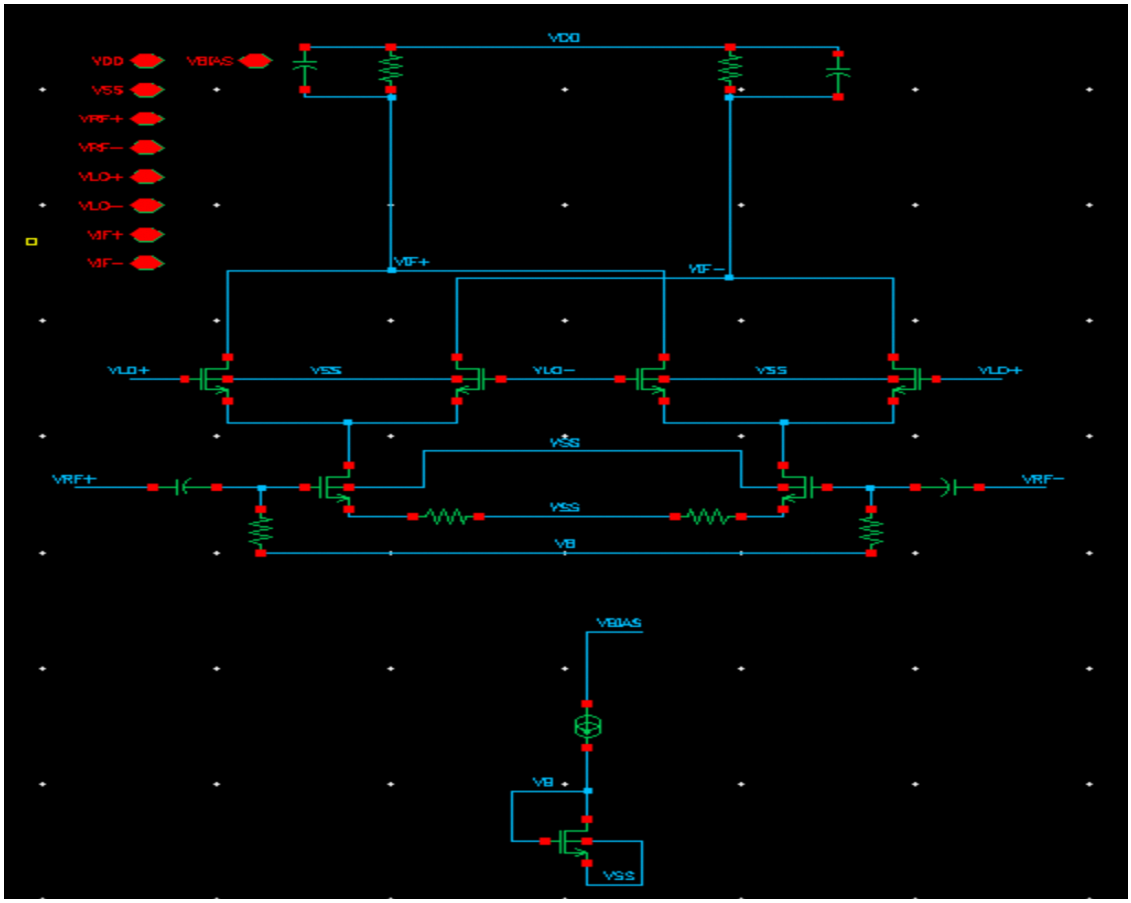
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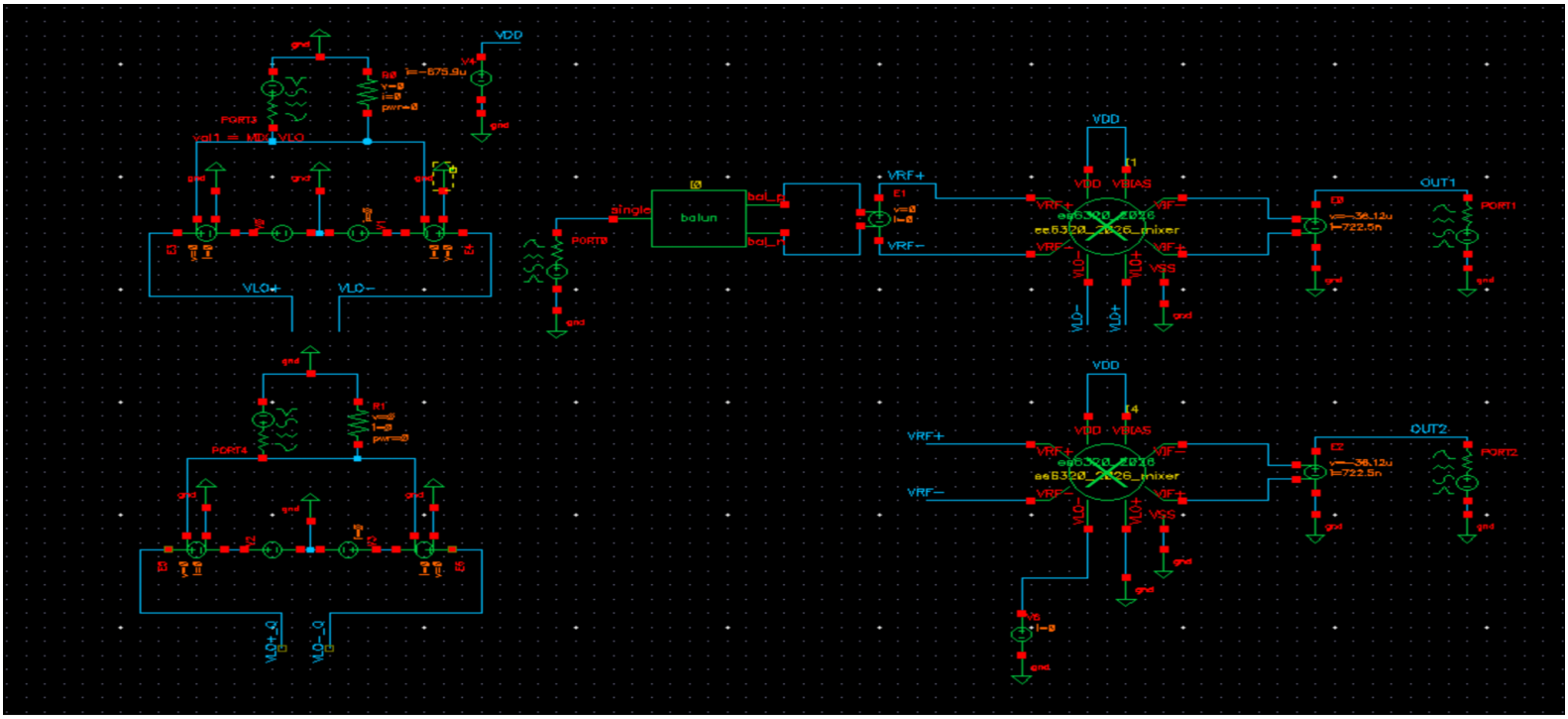
LNA + Mixer Performance Summary Table

	Design Metric	LNA	Mixer	Cascade(-9db bcz of no Balun)	
				Expected	Simulated
Conversion Gain	$f_{IN} = f_{LO}$, $f_{LO} = 3.5\ GHz$	*31.47 dB*	*20.26 dB*	*42.7 dB*	*39.017 dB*
	$f_{IN} = f_{LO} + 10MHz$, $f_{LO} = 3.5\ GHz$	*31.44 dB*	*20.21 dB*	*42.6 dB*	*38.935 dB*
	$f_{IN} = f_{LO} - 10MHz$, $f_{LO} = 3.5\ GHz$	*31.37 dB*	*20.23 dB*	*42.6 dB*	*38.934 dB*
Simulated gain is off by 3db because of mismatches between mixer and LNA and other non idealities					
Noise Figure	$f_{IN} = f_{LO}$, $f_{LO} = 3.5\ GHz$	*1.836 dB10*	*9.375 dB10*	2.81dB10	*2.22 dB10*
	$f_{IN} = f_{LO} + 10MHz$, $f_{LO} = 3.5\ GHz$	*1.84 dB10*	*9.59 dB10*	2.81dB10	*2.25 dB10*
	$f_{IN} = f_{LO} - 10MHz$, $f_{LO} = 3.5\ GHz$	*1.832 dB10*	*9.601 dB10*	2.81dB10	*2.249 dB10*
Linearity IIP3	Input power used for extrapolation	-45 dBm	-50 dBm	-	-50 dBm
	Power of Fundamental Tone at output (at chosen input power)	-13.52 dBm	-29.75 dBm	-	-12 dBm
	Power of IM_3 Tone at output (at chosen input power)	-83.56 dBm	-121.75 dBm	-	-56 dBm
	Extrapolated IIP_3	-9.97 dBm	-4.0029 dBm	-	-26.57 dBm
Power	Total power consumption [Excluding Bias]	525.36 micro W	782 micro W	1.307 mW	1.401 mW
	Bias circuit power consumption	9.6 uW	28.8 uW	38.4 uW	40.3 uW

Mixer Schematic



Mixer Testbench



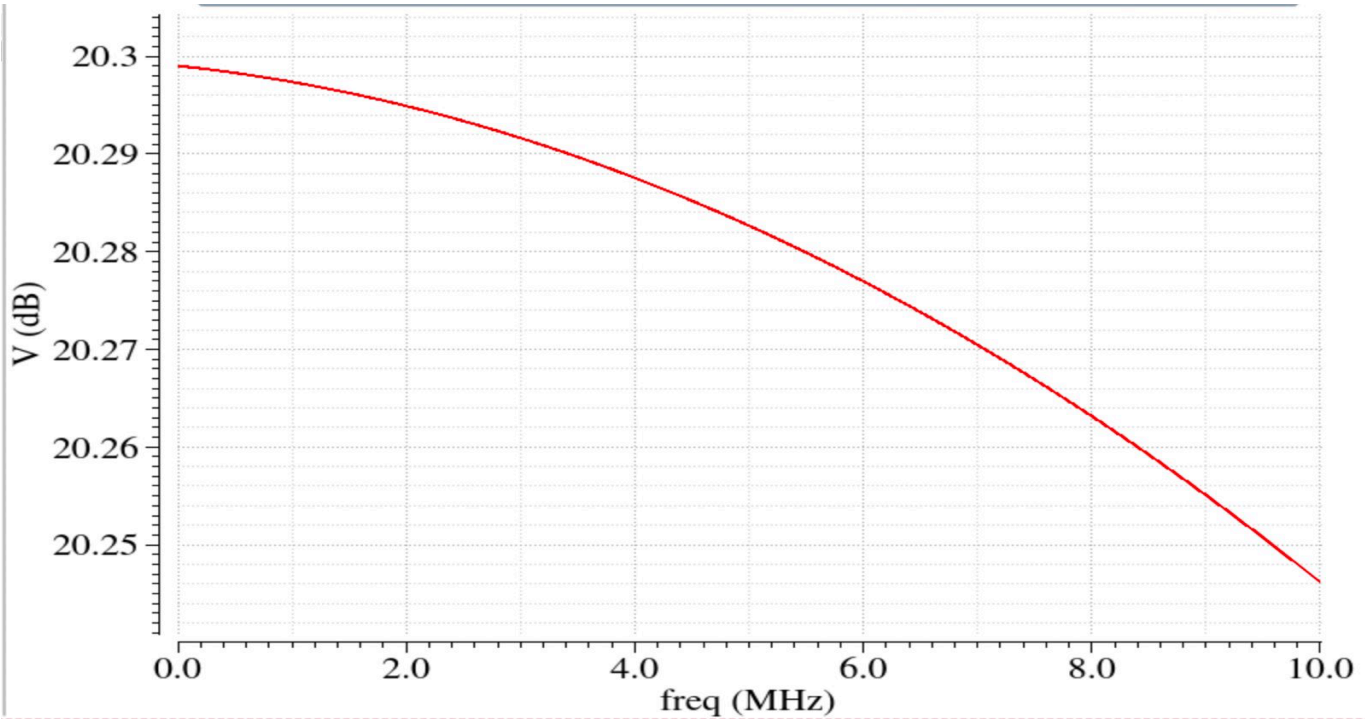
Design Variable Values

Design Variable	Value
Resistance (load, each side)	2.15K ohm
Length of all MOS	120n
Width of Switch MOS (all 4)	75 micro
Width of Transconductance MOS (both MOS)	20 micro
Bias Current	12 micro for mirror and 162u through each transconductance fet
V_LO Amplitude	650 milli Volts

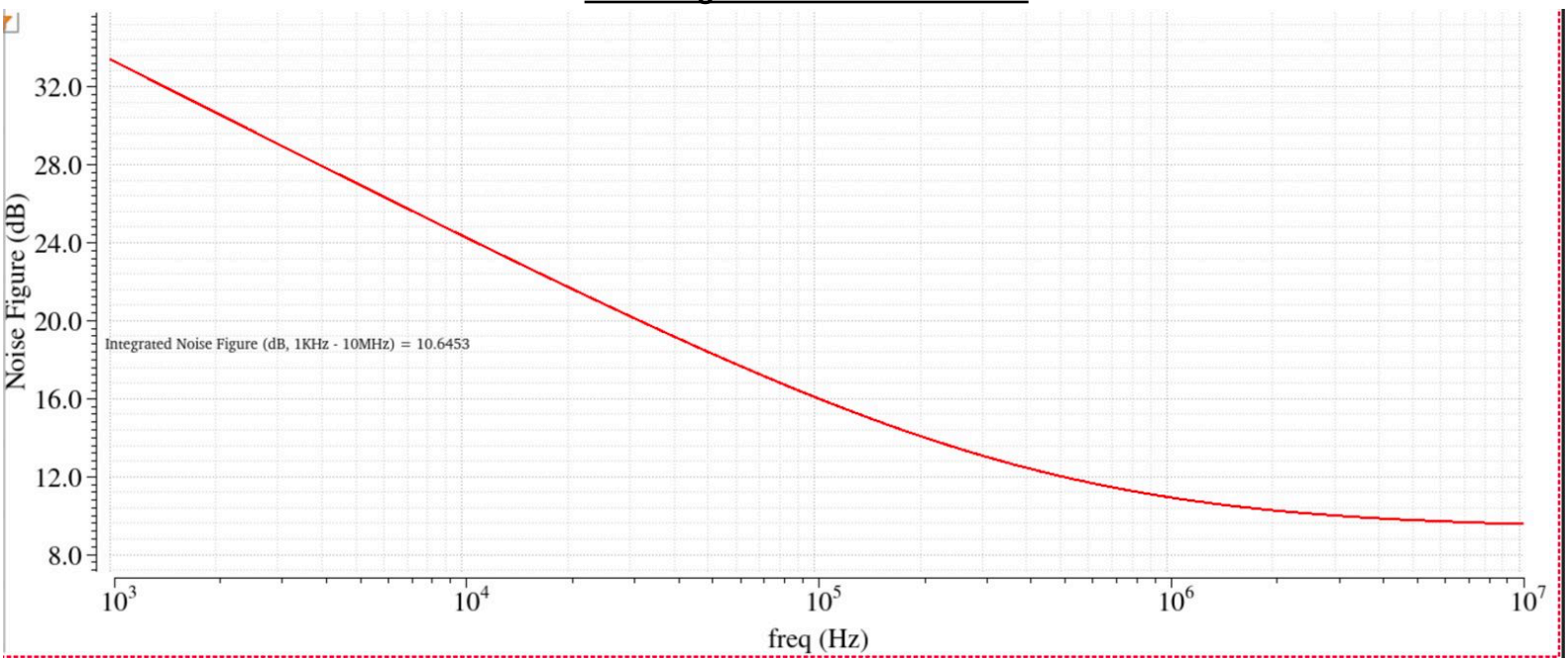
Fixed Constant Parameters

- Transconductance MOS C_gs: 29.92f F
- Current Mirror MOS: W = 1 micro, L = 120n
- Coupling capacitances: 10 pF(for each transconductor fet)
- Bias Resistances: 10K ohm (for each transconductor fet)

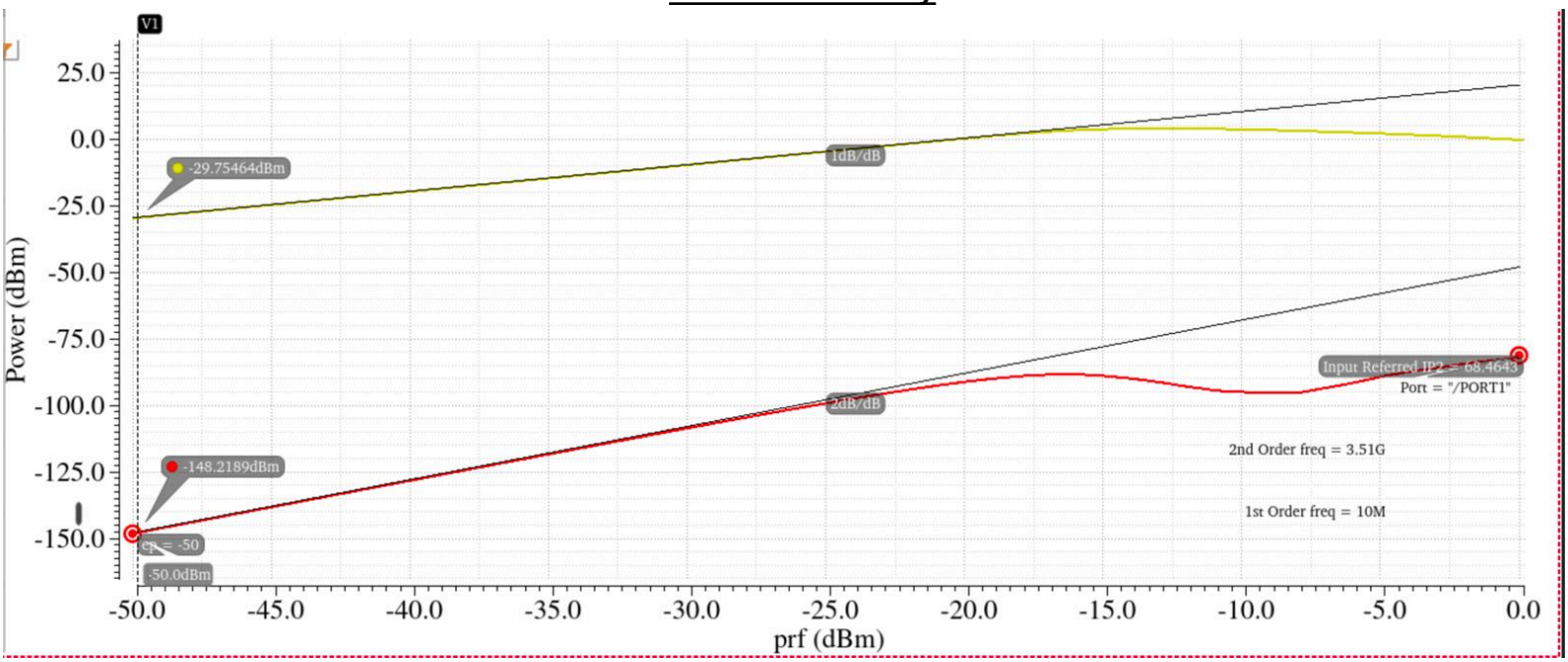
Conversion Gain of Mixer at 3.5 GHz



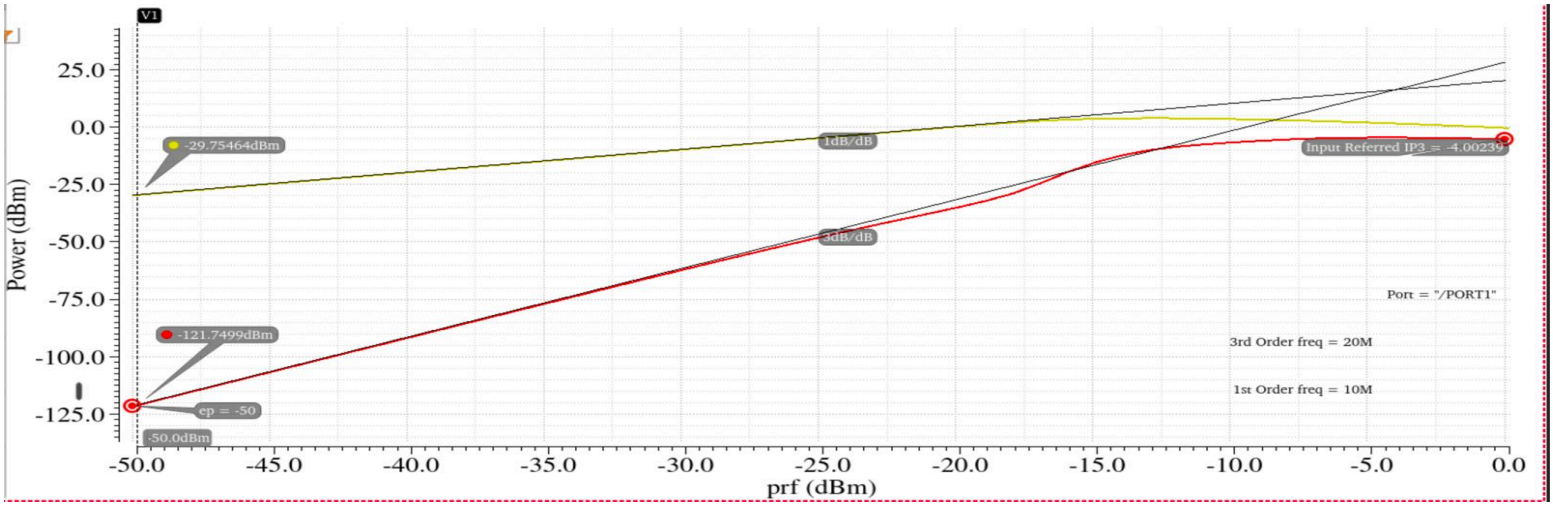
Noise Figure of Mixer at 3.5 GHz



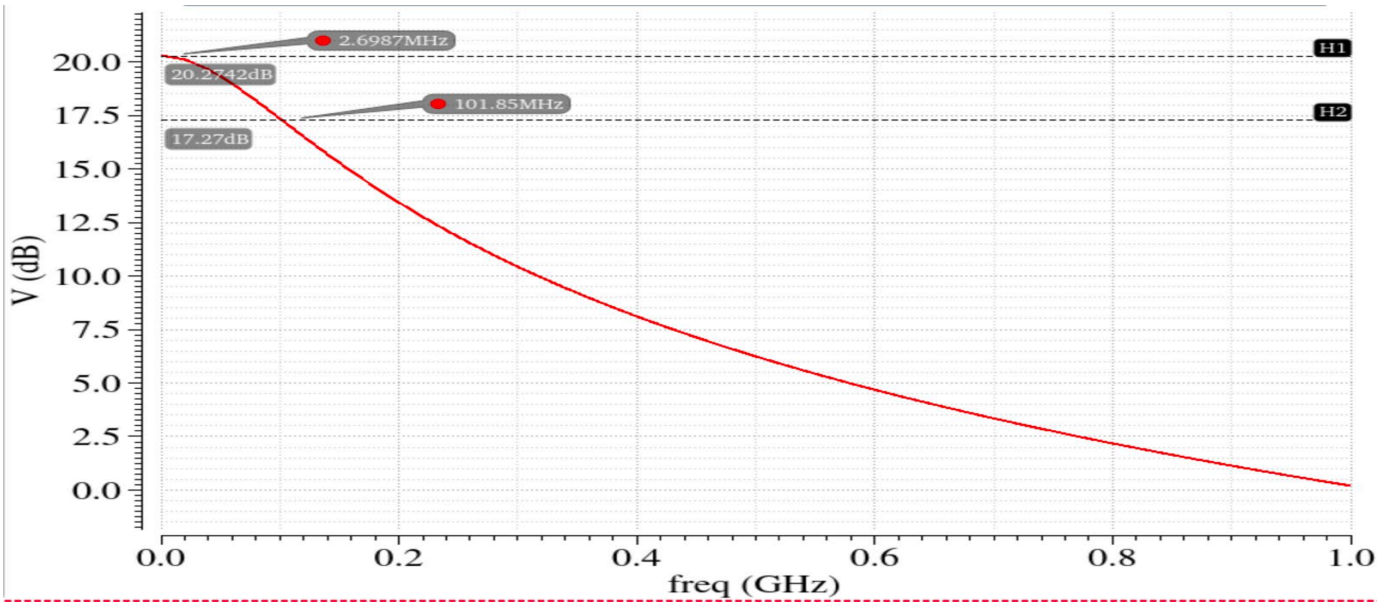
Mixer IIP2 Linearity



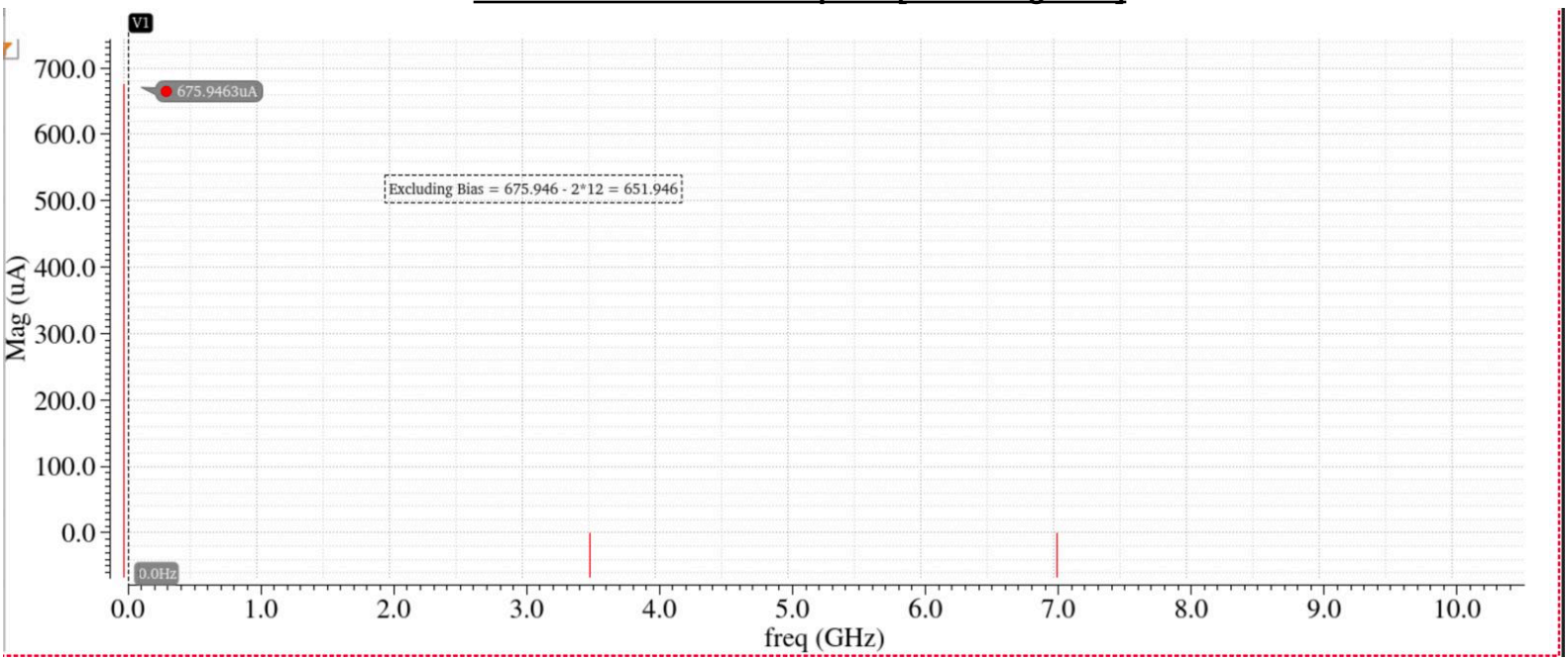
Mixer IIP3 Linearity



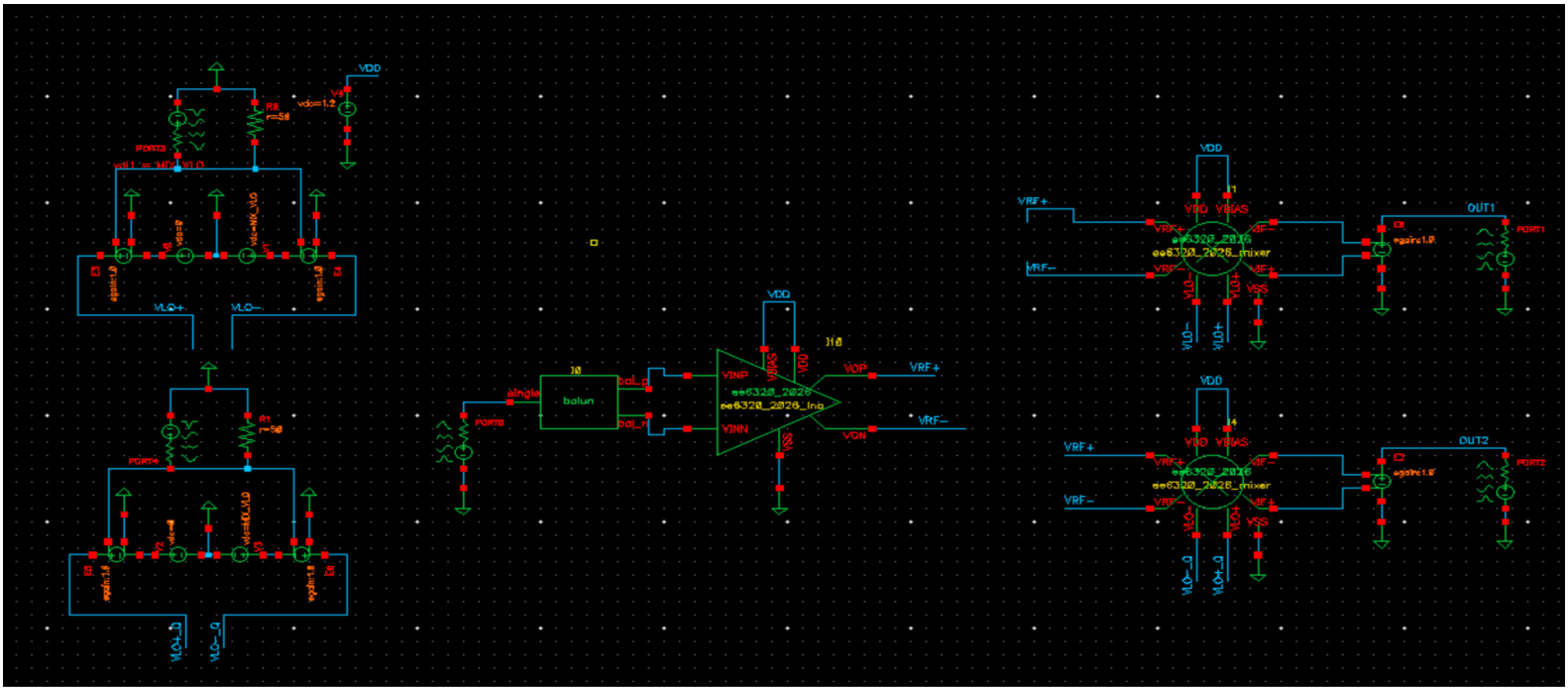
Mixer 3-dB Bandwidth



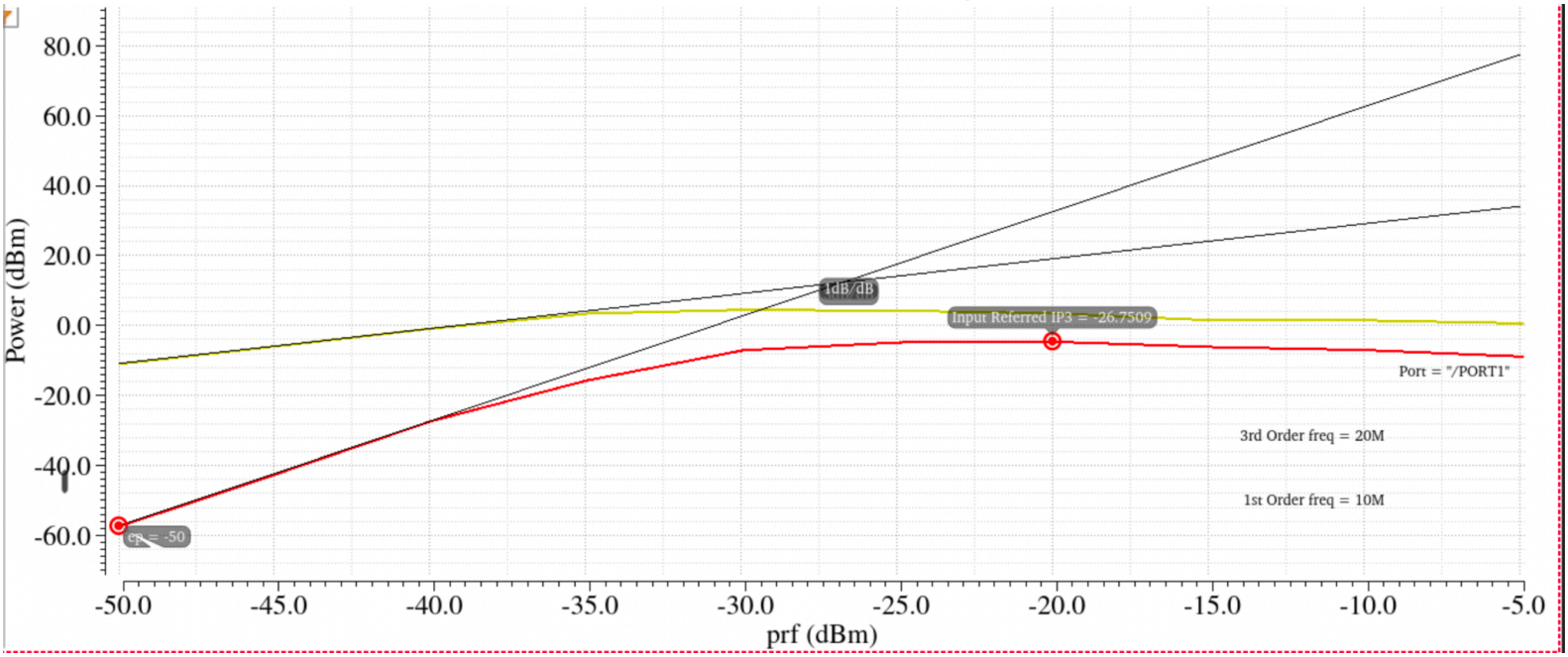
Mixer DC Power Consumption [Excluding Bias]



LNA + Mixer Testbench

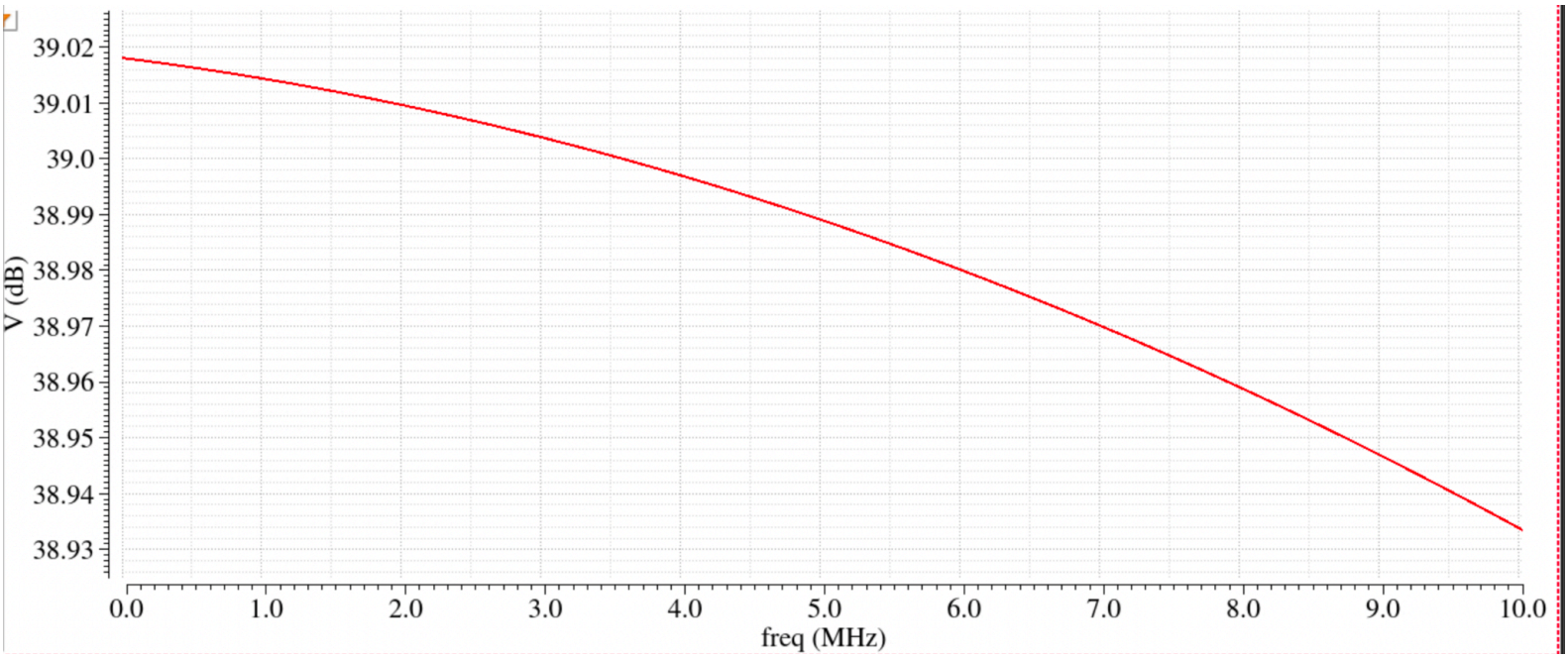


LNA + Mixer IIP3 Linearity

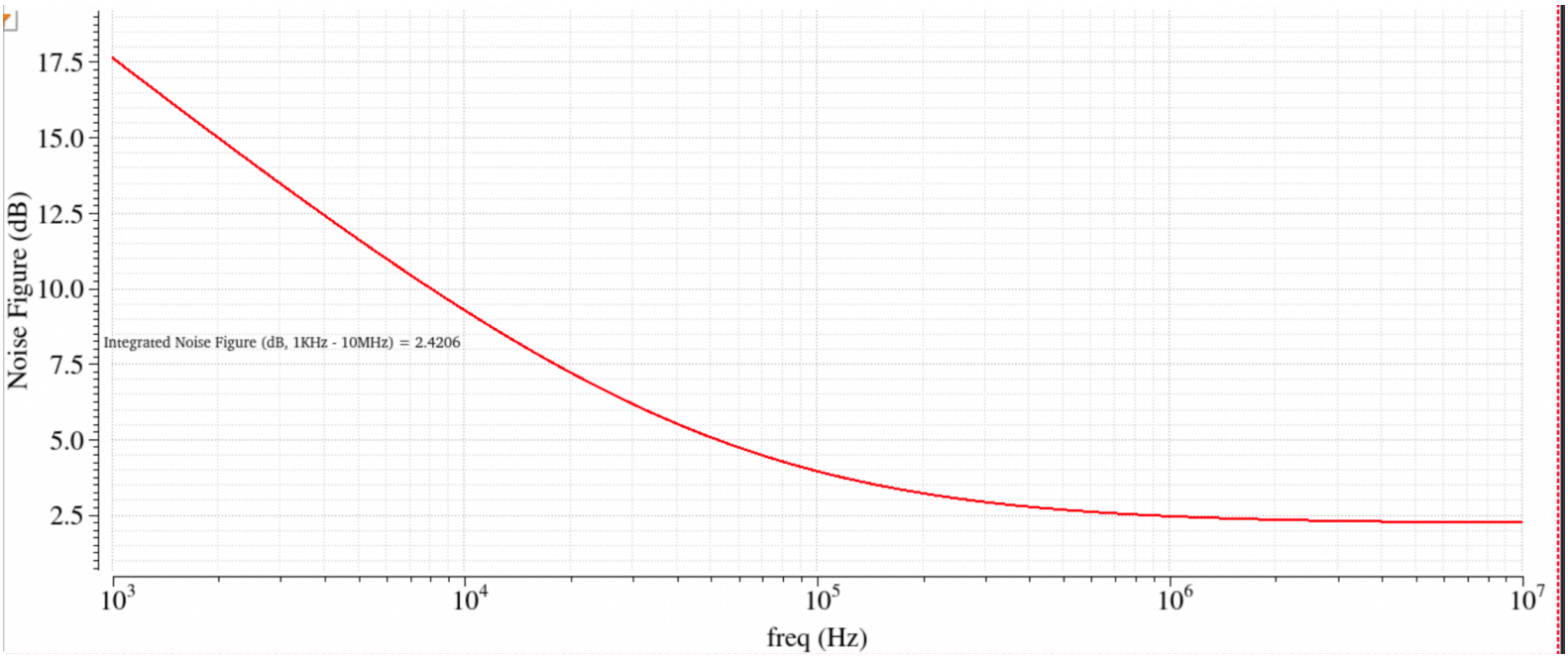


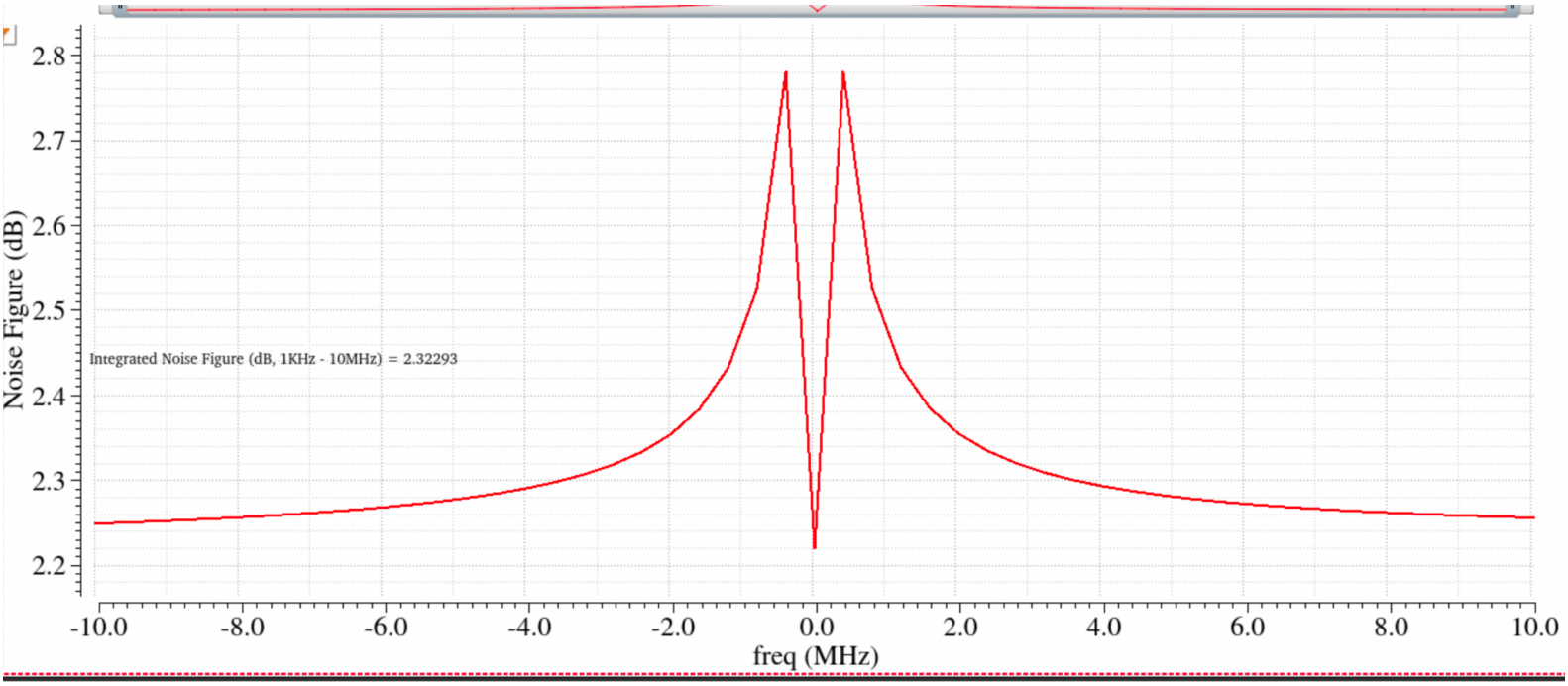
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LNA + Mixer Conversion Gain at 3.5 GHz

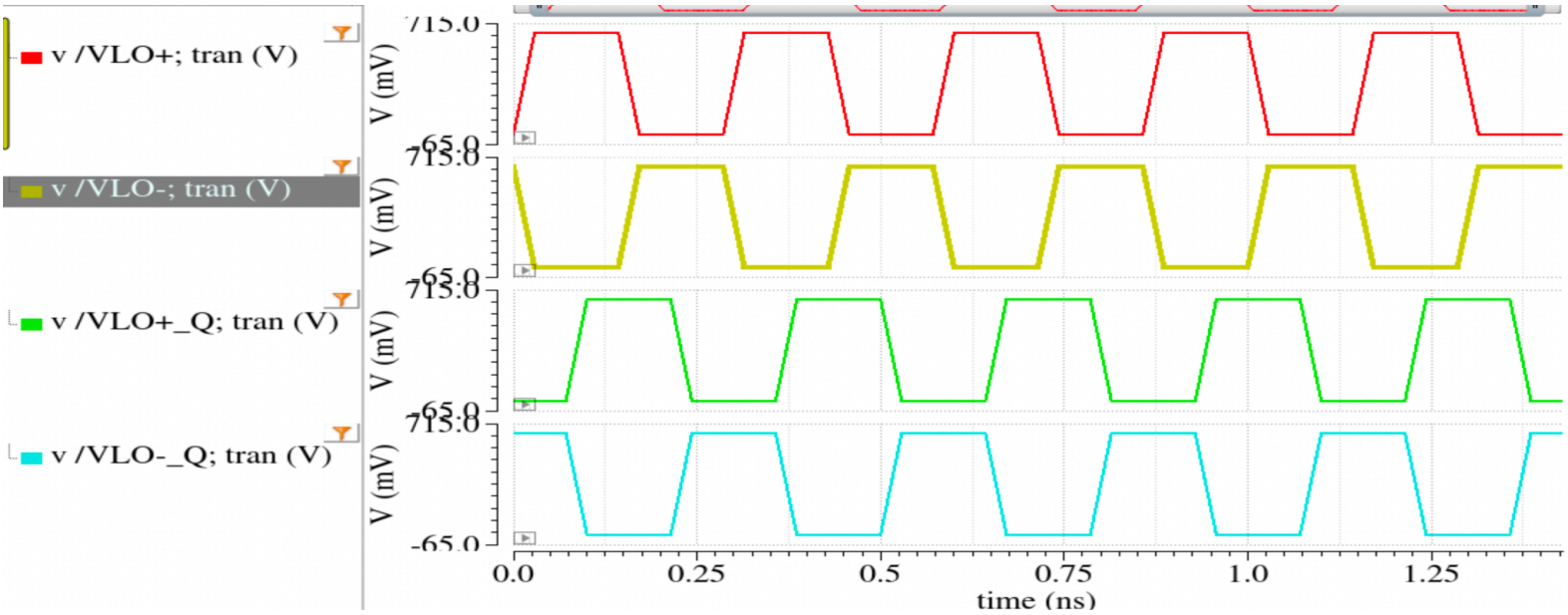


LNA + Mixer Noise Figure at 3.5 GHz





LO Waveform Characteristics



Characteristics taken are *(for generalized manner, as flo varies for various analysis)***:**

- **Frequency:** 3.5GHz = flo
- **Time period:** 1/(3.5 GHz)
- **Delay time:** 0 **(for I Lo signal),** 0.25/(3.5 GHz) **(for Q Lo signal)**
- **Pulse Width:** 0.40/(3.5 GHz)
- **Rise/Fall Time:** 0.1/(3.5 GHz)

Procedure adopted for the Project

- I started with a width of 20u and found a load resistor using required gain. Starting with a Vlo voltage of 1.2V I tweaked Load resistor value and Vlo such that I reach the edge of given gain and NF spec.
- Tried using current bleeders and degeneration resistor for better IIp3 but it started tanking NF. So I tried to increase cascode width for the same bias current so that I get a lower Vdsat for better swing which made IIP3 better.
- After that I tried decreasing bias current as much as possible so that all the given specs meet.

Path to the Project Files

Mixer schematic:
/home/ee23b114/cadence/EE6320_2026/ee6320_2026/ee6320_2026_mixer

Mixer testbench: /home/ee23b114/cadence/EE6320_2026/ee6320_2026/Test2

Mixer_LNA testbench : /home/ee23b114/cadence/EE6320_2026/demo/LNA_MIXER

Calculations done for the Project :

⇒ Hand Calculations for Mixer:

Voltage gain $= \frac{2}{\pi} g_m R_L$

Needed gain = 16db but balun provides 9db so 11db needed

$10^{11/20} = \frac{2}{\pi} g_m R_L$

For $g_m = 2.3mV$ I got resistance of 2.5kΩ

⇒ Starting from these values and $V_{LO} = 1.2V$, I started tweaking such that I am at edge of given specs gain of 20db and NF of < 12db so that IIP3 is good.

got to $R_L = 2.15k\Omega$, current = 162uA through RF fet

for RF fet width = 20u and switching fet = 75u

For low Vdsat higher swing,

For $\omega_{3db} = 100MHz = \frac{1}{2\pi RC}$

⇒ $C = 723fF$

⇒ Cascaded gain = LNA + Mixer − 9 (↳ balun)

= 31 + 20 − 9 = 42db

Cascaded NF = $F_1 + \frac{F_2 - 1}{A}$

$F_1 = 1.83$

$F_2 = 9.375$

$A = 20db = 100$

$\therefore F = 1.83 + \frac{9.375 - 1}{100}$

$$F = 1.91$$

$$NF_{net} = 2.81$$

$\Rightarrow$ IIP_3 Calculation:-

$$\frac{1}{IIP_{3net}} = \frac{1}{IIP_{3LNA}} + \frac{A_{LNA}}{IIP_{3Mixer}}$$

IIP_3 of LNA $= -9.97dbm = 0.1\,mW$

IIP_3 of Mixer $= -4.0096\,dbm = 0.397\,mW$

gain of LNA $= 3db = 35.48$

$$\therefore \text{Cascaded } IIP_3 = \frac{1}{\frac{1}{0.1} + \frac{35.48}{0.397}}$$

$$= 1 \times 10^{-2}\,mW$$

$$IIP_3 = 10 \log \frac{10^{-2}}{10} = -20dbm$$